

Varshitha Thilak Kumar

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Skills

- Computational Thinking & Problem Solving
- ML and basics of DL
- C and C++ Programming
- Data Structures & Algorithms (DSA)
- Java (Basics)
- MATLAB(Basics)
- Python Programming
- Web Development
- Languages Known:English,Tamil,Telugu,Hindi

Projects

1. Canteen Management System (C Language)

- Developed a C-based application for canteen inventory and order management.
- Strengthened C programming skills.
- Faced challenges in implementing inventory management.

2. Text to Morse Code Converter (Java)

- Designed a Java-based text-to-Morse code converter.
- Encountered challenges in configuring sliders for volume and speed adjustments.

3. Automatic-Forklift-Prototype using Pygame and Raspberry Pi 4B

- Developed a smart forklift prototype using Raspberry Pi for real-time control and sensor data processing.
- Simulated operational dynamics using Pygame and OpenGL to visualize mechanical stress and enhance predictive maintenance.

4. DDoS detection in SDN using ML

- Developed a DDoS detection and mitigation system in SDN using Mininet, integrating ML models (Random Forest, SVM) with a RYU controller for real-time traffic classification.
- Implemented automated blocking and rerouting strategies to enhance network resilience against evolving threats.

5. Eater Ease Online Restaurant - Responsive Web Page(UI)

- Built a fully responsive online restaurant website.
- Initially struggled with JavaScript but gained proficiency.

6. Time-Frequency Analysis on PQD using Advanced Signal Processing Techniques (MATLAB)

- Applied advanced signal processing techniques for power quality disturbance analysis.

- Faced difficulty in understanding the mathematical formulas for signal processing methods.

7. Statistical Analysis Project (Mathematics)

- Explored Graphical Lasso and ADMM (Alternating Direction Method of Multipliers) for statistical data analysis.
- Faced challenges in applying ADMM but later focused on alternative computational methods. Conducted mathematical derivations and graphical representations for statistical models.

8. Phishing Website Detection (Machine Learning)

- Developed a phishing website detection system using machine learning models.
- Analyzed a dataset containing legitimate and phishing websites.
- Improved feature extraction techniques to enhance classification accuracy.

Education

Maharishi Vidya Mandir / 10th Graduation

June 2020 - March 2021, Chennai

Passed out with 93.6%

Velammal Vidyalaya / 12th Graduation

June 2021 - March 2023, Chennai

Passed out with 87.2%

Amrita Vishwa Vidyapeetham / B-Tech CSE-AI

August 2023 - July 2027, Coimbatore

Currently pursuing 3rd Year

CGPA: 8.47 SGPA:8.97

Awards and Certifications

- Presented Paper at IEEE Conference (ETIS, Trivandrum - 2024)
Topic: Time-Frequency Analysis on PQD using Advanced Signal Processing Techniques (MATLAB)
Published on IEEE Xplore-<https://ieeexplore.ieee.org/document/10961578>
- **Finalist, Cubicle Code 5.0 Hackathon, 2025** — Conducted by Geek Rooms in collaboration with Pathway and TRAE at Microsoft, Bangalore
- **Finalist in GenAI Hackathon (Value Health Org) - 2024**
Developed GenJournals, an AI-powered journaling application.
- Completed Full Stack Development course (MERN stack) at IITM research park through Copter Code
- Completed Gen AI Course(Vertex AI, Gemini APIs, Imagen, Streamlit and Multimodal RAG) by Gen AI Academy that helped be gain foundational knowledge and skills using Google's Gen AI tools and Platforms
- Oracle Cloud Infrastructure 2025 Certified Foundation Associate
- Certified in C and C++ (FIIT)

